Self-Supervised Learning

Final Examination — Variant B

Reconstructed from photographs for personal study review

Important note. This document was reconstructed from angled, partly blurred, upside-down, and partially obstructed phone photos of a physical exam paper. Text marked in **red** indicates spots where the wording or an option could not be read with confidence. Treat this as a best-effort study aid, not a verified transcript — cross-check against the original paper where it matters. Questions 1–6 were not visible in any of the provided photos.

Part I — Multiple Choice

Question 7

[2 points]

A diagonal-Gaussian VAE encoder returns $\mu = (1, 0)$ and $\log \sigma^2 = (0, \log 4)$. For one sample, it draws $\epsilon = (2, -1)$. It uses the standard reparameterization trick. For reference, the KL divergence can be given as

$$\mathrm{KL}\big(q(z \mid x) \,\|\, \mathcal{N}(0, I)\big) = \tfrac{1}{2} \sum_j \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2\right).$$

What are the sampled z and $\mathrm{KL}(q(z \mid x) \,\|\, \mathcal{N}(0, I))$?

A. $z = (3, -2)$ and $\mathrm{KL} = 2 + \log 2$

B. $z = (2, -1)$ and $\mathrm{KL} = \log 2$

C. $z = (3, -4)$ and $\mathrm{KL} = 2 - \log 2$

D. $z = (3, -2)$ and $\mathrm{KL} = 2 - \log 2$

Question 8

[2 points]

A team trains identical VAEs on 64×64 and 128×128 images with

$$\mathcal{L} = \sum_{p=1}^{HW} \ell_{\mathrm{rec},p} \; + \; \beta \cdot \mathrm{KL}\big(q(z \mid x) \,\|\, p(z)\big),$$

where p indexes all HW pixels and $\ell_{\mathrm{rec},p}$ is the reconstruction loss at pixel p. The above loss is subsequently averaged over the batch. Per-pixel reconstruction errors and latent statistics are initially comparable. If this pixel-summed loss is retained, what change best preserves the reconstruction-regularization trade-off at the higher resolution?

A. Leave β unchanged, because the final batch mean normalizes both the pixel and latent dimensions.

B. Divide the KL term by four, because a larger input makes the prior four times more informative.

C. Multiply β by four, because the reconstruction sum has about four times as many terms while the latent KL has not changed dimension.

D. Divide β by four, because each high-resolution pixel should receive less regularization.

Question 9

[2 points]

An MAE splits an image into 196 patches and masks 75% of them. Its 12-layer encoder produces embeddings for the visible patches, while the 4-layer decoder aims to restore the masked image (ignore the class token). Assume attention-score cost per layer is proportional to the square of the token count, and embedding widths are equal. Which pair is correct: decoder token count and total attention-score cost relative to the reference (12-layer encoder) cost?

A. 196 tokens and 19/48 of the reference cost.

B. 49 tokens and 19/16 of the reference cost.

C. 49 tokens and 1/16 of the reference cost.

D. 196 tokens and 5/16 of the reference cost.

Question 10

[2 points]

A Gaussian Kernel Density Estimator is fitted to one-dimensional samples clustered in two narrow groups around $-J$ and 4 (bandwidth h). What is the most likely consequence when the bandwidth becomes very large?

A. [unclear — text partially cropped/upside-down in photo]

B. [unclear]

C. [unclear]

D. The two modes may merge into one broad mode, assigning substantial density between the observed groups.

Note: This question was photographed upside-down and partly cropped; option D is the clearest, others could not be fully read.

Question 11

[2 points]

An affine coupling layer splits $x = (x_a, x_b)$ and computes $y_b = x_b \odot \exp(s(x_a)) + t(x_a)$ and $y_a = x_a$. What is the inverse transformation?

A. $x_b = (y_b - t(y_a)) \odot \exp(-s(y_a)),\ \ x_a = y_a$

B. $x_b = (y_b + t(y_a)) \odot \exp(s(y_a)),\ \ x_a = y_a$

C. [wording partially unclear in photo]

D. [wording partially unclear in photo]

Note: Options A/B reconstructed from partially legible upside-down text; C/D could not be confidently read.

Question 12

[2 points]

A PixelCNN-like autoregressive system receives an input sketch c and generates $G(c)$, but the discriminator sees only the output image and never the sketch. What important capability is hampered by this design?

A. The discriminator can no longer evaluate whether a realistic-looking output is consistent with the input sketch.

B. The generator loses the ability to produce colour images.

C. The generator cannot evaluate an exact conditional density estimator.

D. The system can no longer be trained adversarially.

Question 13

[2 points]

During GAN training, the discriminator improves very quickly, reaching nearly perfect accuracy, while the generator's samples are still poor. Which explanation is most plausible?

A. The generator should now be removed, because the discriminator alone can generate new samples.

B. Perfect discriminator accuracy guarantees that the generator gradients have the largest possible signal.

C. The generator has already learned the complete data distribution, because the discriminator certifies sample quality.

D. The discriminator has become too strong, so the generator receives little useful learning signal / gradient.

Question 14

[2 points]

You train a rectified-flow model $f_\theta(x_t, t)$. Due to an implementation error, the network is always given $t = 1$, although x_t is constructed using different values of t. What is the most likely consequence?

A. The target velocity $v = z - x$ becomes zero for every training example.

B. Every x_t automatically becomes the pure noise sample z.

C. There is no effect because x_t always reveals the exact value of t.

D. The network receives incorrect time information, so its velocity predictions across the path are likely to be less accurate.

Question 15

[2 points]

A network is pretrained to predict the permutation used to create a jigsaw puzzle. A linear probe on a middle feature block outperforms a probe on the final block. Which explanation is most plausible?

A. A linear probe updates middle feature blocks but does not update final feature blocks.

B. The middle block preserves finer spatial detail, so it must be more useful than the final block for every downstream task.

C. Middle feature blocks always contain more parameters than final blocks and are therefore more informative.

D. The final block may specialize to the jigsaw-prediction task and retain less linearly accessible semantic information.

Question 16

[2 points]

For two self-supervised learning methods A and B, Method A achieves higher linear-probe accuracy than Method B. After fine-tuning on the same downstream task, Method B achieves higher accuracy. For which application is Method B likely the better choice?

A. Visualizing the frozen features using dimensionality reduction.

B. Clustering unlabeled data in feature space.

C. Self-supervised pretraining to initialize supervised training on a small set of labeled data.

D. Using the frozen pretrained representation with a linear classifier.

Question 17

[2 points]

A self-supervised pretext task predicts whether an image was rotated by 0°, 90°, 180°, or 270°. For which type of data is this pretext task likely to be least useful?

A. Portrait photographs of people.

B. Images from the dashcam of a car.

C. Birds-eye-view / top-down images from satellites.

D. Scanned documents.

Question 18

[2 points]

SimSiam minimizes the symmetric objective $D(p_1, \text{stopgrad}(z_2))$ between a shared encoder and predictor. Which is correct?

A. Both terms compute the same objective.

B. The first term freezes the shared encoder and predictor for the complete objective.

C. Stop-gradient updates only the encoder evaluated on view 1, while the second encoder is updated through the shared/differentiable path.

D. In each term the stop-gradient is applied to one branch, so one encoder is detached across the two terms, and one differentiable path is used, so the encoder does not collapse to a trivial solution.

Note: This page was photographed upside-down and the exact option wording is partly reconstructed; verify against the original.

Question 19

[2 points]

Why is the EMA teacher in DINO used (rather than a hard target)?

A. The EMA teacher accumulates representations from previous batches and reuses them as a large collection of negative examples.

B. The student matches the outputs of a slowly-updated EMA teacher; assigns almost every image as a negative.

C. Two DINO runs without a teacher would produce different images in the batch as negative examples.

D. The multi-crop strategy converts the images in a small batch into a large collection of positive pairs.

Note: Wording partially reconstructed from garbled/upside-down text — verify against the original.

Question 20

[2 points]

Two DINO runs, Run A and Run B, differ only in the role of centering and sharpening (teacher temperature). In Run A, the teacher assigns almost uniform probability over all outputs (a collapse failure). In Run B, the teacher output is nearly one-hot / concentrated on the same coordinate regardless of input. Based on the role of centering and sharpening, what is the most likely cause and correction for each run?

A. Run A: raise the teacher temperature; Run B: lower the teacher temperature.

B. Run A: remove local crops matched to each other.

C. Run A: restore correct running-center subtraction; Run B: stop updating the EMA teacher permanently.

D. Run A: add more local crops to the teacher; Run B: remove all global crops from the student.

Note: Question stem is condensed/reconstructed from a badly angled photo — the exact framing of Run A vs Run B may not be precise.

Question 21

[2 points]

Which statement best describes Noisy Student training?

A. A student learns to match the prediction of a teacher, but with noise applied to its input.

B. A teacher is trained with random labels so that the student becomes robust to incorrect supervision.

C. A teacher produces pseudo-labels for unlabeled data, and a student learns to reproduce the teacher's predictions without augmentation.

D. A student generates labels for the training data, and a teacher learns to reproduce the student's loss through both networks.

Question 22

[2 points]

CycleGAN translates summer road scenes into winter road scenes using unpaired data. Target-domain images look realistic and round-trip cycle error is low, yet lane boundaries and small traffic signs sometimes move, causing a downstream pixel error to fail. Which intervention most directly addresses the observed failure?

A. Treat the low cycle error as proof that the controller, rather than the translation, must be wrong.

B. Delete the reverse generator, because cycle consistency is the cause of any low round-trip error.

C. Increase only discriminator capacity, because a perfect target-domain discriminator guarantees that each source object's position is preserved.

D. Condition the discriminator on a semantic segmentation of the original image.

Question 23

[2 points]

How are the parameters of an EMA teacher network updated?

A. The teacher parameters are updated directly by backpropagating the student's loss through both networks.

B. The teacher parameters are updated as a moving average of the student parameters throughout training.

C. The teacher remains within a specified bound from its initial parameters.

D. The teacher is reinitialized from random parameters after every training epoch.

Question 24

[2 points]

You have developed a system to provide pseudo-labels for traffic-sign images and train a network on this data using strong augmentations. Horizontal flipping changes a left-turn sign into a right-turn sign, but the original pseudo-label is retained. What is the most appropriate correction?

A. Generate pseudo-labels from strongly corrupted images and train the network only on clean images.

B. Remove the genuinely labeled examples so that the conflicting pseudo-labels dominate training.

C. Increase the probability of horizontal flipping so that the network becomes invariant to sign direction.

D. Use only label-preserving augmentations, or update the target consistently when an augmentation changes the sign's meaning.

Question 25

[2 points]

Two segmentation models agree on 97% of unlabelled pixels but both perform poorly on a small labeled validation set. Which conclusion is best supported?

A. The labeled validation set should be ignored because it is not part of training.

B. The models may have converged to the same incorrect solution and are reinforcing each other's errors.

C. At least one model must have errors, but it is not possible to tell which from agreement alone.

D. Their agreement proves that the shared predictions are correct.

Note: Option C wording is uncertain — the photo of this option was upside-down and partly cut off.

Part II — Detailed Questions

Answer each question in the space provided. Show your reasoning, pseudo-code, and calculations where necessary.

Question 26

You are training a variational autoencoder (VAE) to reconstruct CIFAR-10 images.
(a) Write pseudocode for the VAE's `encode(x)`, `decode(z)`, and `forward(x)` methods; the forward pass must sample the latent code using the reparameterization trick. You may use the following commands in your pseudocode: `net(in_dim, out_dim)` creates a feedforward neural network block with the given input and output dimensions; `linear(in_dim, out_dim)` creates a linear projection with the given input and output dimensions; `rand(dim)` samples a random vector of given size from a unit normal distribution. *[10 points]*

```
def encode(x):
    ...

def decode(z):
    ...

def forward(x):
    ...
    z = ...
    return decode(z)
```

(b) Briefly explain why the reparameterization trick is required when training a VAE, and what problem it solves. *[points not clearly legible]*

Question 27

[part totals not fully legible]
You are given 50,000 unlabeled indoor images and 50 labelled validation images. The labelled images may be used for evaluation and limited cluster interpretation, but not for ordinary supervised training.
A pretrained encoder produces a feature vector z_i for every image. The images contain approximately six semantic object categories, but visual conditions vary strongly because of lighting, scale, pose, and background.
(a) Describe briefly two preprocessing operations that you would consider before clustering the features. For each operation, state what it does, what problem it may solve, and one possible disadvantage. *[4 points]*
(b) Compare the suitability of k-means, DBSCAN, and overclustering with k-means (for example $k = 12$), when each semantic category may contain several visual modes. State which strategy you would initially choose and justify your decision briefly. *[4 points]*
(Parts (c) and (d) of this question were not photographed / not visible in the provided images.)

Question 28

[10 points total]
You are given paired examples (x_i, y_i), where x_i is a semantic segmentation map of an indoor scene and y_i is the corresponding RGB image. The goal is to generate realistic RGB scenes that preserve the locations and boundaries specified in the segmentation map. You are considering

KDE, a **normalising flow**, an **unconditional GAN**, and a **conditional GAN**.
(a) Compare the four model families for this task. Explain why KDE is difficult to apply directly to images, why an unconditional GAN is insufficient, one advantage and one disadvantage of a normalising flow, and why a conditional GAN is a natural main choice. *[4 points]*
(b) A colleague implements the following training procedure. Here, G is the generator mapping (x, z) to an RGB image, D is the discriminator that should output 0 for a real image and 1 for a fake, and z is a noise vector used to introduce stochastic variation:

procedure TRAIN
  for each batch (x, y) **do**
    $z \leftarrow z_{\text{fixed}}$
    $y_{\text{fake}} \leftarrow G(x, z)$
    $\mathcal{L}_D \leftarrow -\log D(y) - \log(1 - D(y_{\text{fake}}))$
    Update D using $\mathcal{L}_D$
    $y_{\text{fake}} \leftarrow \text{StopGradient}(G(x, z))$
    $\mathcal{L}_G \leftarrow -\log D(y_{\text{fake}})$
    Update G using $\mathcal{L}_G$
  end for
end procedure

After training, generated images sometimes look realistic but do not follow the input segmentation map, changing the noise at test time has little effect, and the generator improves very slowly. Identify three errors or weaknesses in the procedure. For each one, explain the failure it may cause and state how the procedure should be corrected. *[6 points]*

Question 29

[20 points total — exact part breakdown partly reconstructed]
For the City of Rome, you are tasked to develop a system that automatically detects cracks on camera images of bridge pillars and road. They supply you with 50 million unlabeled inspection images plus a web crawl of mixed images of bridge pillars and road. Adjacent video frames are near-duplicates of each other, and most images do not contain any cracks. Unfortunately, the city has no money to pay experts to sort through the data, while the rare images of cracks are near identical. Therefore, you decide to train a DINOv2-style encoder for image-level crack recognition and patch-level crack localization.
(a) Describe DINO multi-crop student-teacher training: views, target formation, gradient flow, teacher update, and center update. *[7 points]*
(b) [Not clearly legible in the photos — appears to ask about masked-patch loss and KoLeo regularization] *[5 points]*
(c) Propose a data curation and sampling pipeline that addresses the problems of exact/near duplicates and long-tail imbalance. Explain how your solution addresses each problem. *[9 points, exact number partly obscured]*
(d) You convince an expert at the university to label a few images, providing you with a small test set. Name two image-level and one patch-level evaluation methods. State for each method what result you expect from a well-working training. *[3 points]*

Question 30

[10 points for part (a)]
Humans can estimate articulation and its associated force just by observing other humans interacting with doors and cabinets.

You collect a dataset that contains many RGB videos of humans interacting with doors and cabinets, but these videos have no force annotations. A much smaller collection of similar objects was recorded with the same video setup, but using specialized equipment to measure the interaction forces. Given this data, develop a method to predict the peak interaction force directly from a human RGB video.

(a) This is an open-ended design question. Propose a complete approach (it can consist of multiple models or steps) that overall addresses the following:

- Select and justify an appropriate self-supervised learning method.
- Describe the main components of the model architecture.
- Explain how the unlabeled human videos and force-labelled gripper recordings are used during training.
- Formulate the complete training objective (for objectives discussed in the lecture, it is sufficient to refer to them by name) and define the purpose of each loss term.
- Explain how the approach supports cross-embodiment transfer and peak-force prediction.

[10 points]

(b) Critically evaluate your approach. State any key assumption and identify one likely failure mode, and describe a small experiment to assess if force-prediction transfers to human video or unseen objects. *[3 points]*

(c) Your colleague asks an AI assistant how DINO could solve this problem. The AI assistant answers: *"The DINO teacher is directly trained using the measured gripper forces. It therefore produces reliable force pseudo-labels for the unlabeled human videos. Matching the student to these labels guarantees that the learned representation is embodiment-invariant."* Identify two technically incorrect or unjustified claims in this answer. *[2 points]*

End of reconstruction. Questions 1–6 were not visible in the provided photos. Some option text on Questions 10, 11, 18–20, and 25, and part (b) of Question 29, could not be read with confidence and are marked above.